LITERATURE REVIEW AND COMPETITOR ANALYSIS

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1. Problem Statement:

The accurate assessment of credit risk—specifically the Probability of Default (PD)—is the quantitative bedrock of modern commercial banking and capital adequacy planning under Basel III and IFRS 9. Historically, retail credit underwriting has relied on generalized linear scorecards (e.g., Logistic Regression). These traditional frameworks are favored by supervisors for their parametric transparency and inherent "monotonicity" (the mathematical guarantee that incremental debt or delinquency never decreases default risk). However, parametric scorecards fail to capture high-dimensional non-linear interactions, tail-risk interdependencies, and latent default dynamics within complex borrower profiles, leading to sub-optimal discrimination and potential underestimation of systemic tail losses.

While advanced non-parametric Machine Learning (ML) architectures—particularly Gradient Boosted Decision Trees (GBDT) such as LightGBM and XGBoost—demonstrate superior predictive discrimination, financial regulators (e.g., the Federal Reserve via Supervisory Guidance SR 11-7) classify these models as non-auditable "Black Boxes". Consequently, lending institutions face an operational impasse: deploy complex ensemble architectures that maximize default classification but risk regulatory rejection and violations of adverse action reporting (Equal Credit Opportunity Act, ECOA), or maintain interpretable linear scorecards at the expense of capital efficiency and predictive accuracy.

This project resolves this accuracy–interpretability trade-off by engineering an end-to-end, leakage-free credit default classification framework. By integrating Synthetic Minority Over-sampling (SMOTE) with a leaf-wise constrained LightGBM classifier, post-hoc Shapley Additive Explanations (SHAP), cost-sensitive threshold calibration, and economic fairness audits, this research provides an empirically validated, supervisor-compliant blueprint for deploying non-linear machine learning in retail lending.

2. Literature Review

2.1. Introduction

Credit risk assessment models serve as the gatekeepers of institutional stability, determining both credit allocation and minimum regulatory capital provisions. In the post-2008 regulatory architecture established by Basel III, supervisory authorities mandated empirical soundness, robust stress-testing, and rigorous probability estimation protocols. Over the past decade, the rapid expansion of consumer credit datasets has driven an academic transition from classical parametric scoring toward non-parametric machine learning. This literature review synthesizes the prevailing scholarship across four interconnected dimensions: the comparative efficacy of statistical baselines versus ensemble architectures, methodologies for resolving extreme class imbalance, the operational viability of Explainable AI (XAI) frameworks, and the quantitative auditing of algorithmic fairness under banking compliance standards.

2.2. Traditional Statistical Baselines vs. Ensemble Machine Learning

The foundational quantitative frameworks of credit assessment were established by Altman (1968), who introduced linear discriminant analysis (Z-score) to evaluate corporate bankruptcy, and later extended through generalized linear models, notably logistic regression. For decades, logistic regression remained the undisputed

industry standard due to its exact coefficient interpretability, numerical convexity, and ease of deployment into standard scorecard formats.

However, benchmark studies over the last decade have systematically demonstrated the structural limitations of linear scoring. In a seminal benchmarking study across dozens of real-world credit datasets, Lessmann et al. (2015) established that non-linear ensemble algorithms—specifically gradient boosting and random forests—statistically outperform standard logistic regression in area under the receiver operating characteristic curve (ROC-AUC) and Brier score metrics. Khandani et al. (2010) corroborated these findings on consumer transaction data, demonstrating that non-parametric machine learning reduces default rates by 6% to 25% while expanding credit approval volumes, primarily because tree-based architectures capture localized feature interactions that linear models smooth over.

Within the gradient boosting ecosystem, recent research has increasingly shifted toward histogram-based algorithms. Ke et al. (2017) formulated LightGBM, introducing Gradient-Based One-Side Sampling (GOSS) and Exclusive Feature Bundling (EFB) alongside a leaf-wise (best-first) tree growth strategy. In contrast to standard level-wise tree growth (employed in classical Random Forests and baseline XGBoost), leaf-wise expansion splits the node that achieves the highest delta loss, resulting in asymmetrical decision trees that more effectively segment concentrated tail-risk distributions. Benchmarking reviews by Dastile et al. (2020) and Aboulmaouda et al. (2026) confirm that LightGBM achieves state-of-the-art discrimination on tabular credit datasets while significantly reducing training wall-clock latency, establishing its viability for large-scale financial applications.

2.3. Class Imbalance and Resampling Dynamics in Credit Scoring

A pervasive methodological challenge in credit risk modeling is extreme class imbalance: real-world retail loan default rates typically range between 1% and 5%. Under such skew, standard classification algorithms suffer from the "accuracy paradox," converging to a trivial local optimum that predicts universal non-default. To force learning on rare events, data-level over-sampling interventions are widely deployed.

The Synthetic Minority Over-sampling Technique (SMOTE), introduced by Chawla et al. (2002), addresses this vulnerability by synthesizing minority samples along feature-space line segments joining k-nearest minority neighbors, rather than duplicating instances. Suhadolnik et al. (2023) demonstrated that combining SMOTE with ensemble decision trees significantly enhances minority-class recall and the Matthews Correlation Coefficient (MCC) in retail credit portfolios. However, recent critical reviews (Dastile et al., 2020; Montevechi et al., 2024) caution that aggressive synthetic oversampling can distort the empirical prior probability distribution, shift the decision boundary into majority-class clusters, and cause severe probability miscalibration. Furthermore, leakage occurs when oversampling is performed across the entire dataset prior to partitioning—a common error documented in applied machine learning literature. Consequently, contemporary methodology demands that resampling be confined exclusively to training folds, accompanied by rigorous out-of-sample probability calibration and decision threshold tuning.

2.4. Post-Hoc Interpretability and Model Risk Governance

Despite the documented predictive superiority of ensemble models, their operational deployment in regulated finance is severely constrained by supervisory guidance, such as the Federal Reserve's SR 11-7 framework

on Model Risk Management. SR 11-7 mandates conceptual soundness, rigorous outcome validation, and detailed process transparency. Furthermore, under the U.S. Equal Credit Opportunity Act (ECOA) and the Fair Credit Reporting Act (FCRA), lenders issuing adverse decisions (e.g., loan rejections) are legally required to provide specific, defensible reasons ("adverse action notices") to the applicant.

Traditional tree-based feature importance metrics (such as Gini impurity decrease or split count) fail to satisfy regulatory standards because they provide only global, uncalibrated rankings devoid of directionality or local attribution. To bridge this gap, Lundberg and Lee (2017) formalized Shapley Additive Explanations (SHAP), a cooperative game-theoretic framework that assigns each feature an additive importance attribution (ϕ_i) reflecting its marginal contribution across all possible feature subsets. For tree-based ensembles, the TreeSHAP formulation computes exact Shapley attributions in polynomial time.

Hadji Misheva et al. (2021) and Pathi and Pothineni (2025) evaluated post-hoc XAI methods across financial risk applications, establishing that SHAP provides greater consistency, missingness satisfaction, and localized stability than local surrogate methods like LIME. Local SHAP force plots allow underwriters to decompose individual margin outputs ($f(x)$) into base values and granular feature-level risk mitigators or amplifiers, directly satisfying adverse action mandates while retaining the non-linear predictive capacity of gradient-boosted trees.

2.5. Algorithmic Fairness and Disparate Impact Auditing

The growing deployment of algorithmic scoring has elevated regulatory scrutiny regarding algorithmic bias and systemic discrimination. Financial regulatory bodies, including the Consumer Financial Protection Bureau (CFPB), enforce strict disparate impact standards: an algorithm may be deemed unlawful under ECOA if it disproportionately denies credit to protected classes—even in the absence of explicit discriminatory intent or protected demographic attributes.

Hardt et al. (2016) established formal mathematical definitions for algorithmic fairness, contrasting demographic parity with equalized odds and equal opportunity (True Positive Rate parity). In credit underwriting, False Positive Rate (FPR) parity is of primary concern: a fair model should not exhibit an elevated "false alarm" rate for vulnerable or disadvantaged groups relative to privileged cohorts. As Alonso and Carbó (2020) and Chen (2024) observe, academic studies frequently examine fairness in the presence of synthetic or protected attributes; however, in actual retail banking, explicit demographic variables (race, gender) are legally prohibited from model training. Hence, rigorous model validation necessitates auditing economic proxy variables (e.g., annual income tiers or neighborhood metrics) to ensure that optimization for minority default detection does not introduce disparate rejection rates.

2.6. Synthesis and Identified Research Gap

Systematic reviews of credit scoring (Aboulmaouda et al., 2026; Montevechi et al., 2024; Shi et al., 2022) reveal a persistent bifurcation in published literature:

1. Machine learning studies frequently optimize global statistical ranking metrics (such as ROC-AUC) in an operational vacuum, often ignoring target leakage caused by post-origination repayment artifacts (e.g., historical collections, paid principal, or recovery balances).
2. Conversely, banking literature focuses heavily on traditional linear scorecards for compliance, treating complex ensembles as un-deployable due to "black box" risks.
3. Furthermore, studies applying SMOTE rarely address the downstream probability distortion that degrades practical decision threshold calibration, and few papers combine data leakage controls, XAI attributions, and proxy fairness auditing into a single reproducible pipeline.

This capstone project directly fills this gap. It synthesizes a strictly at-origination, leakage-free feature architecture, evaluates leaf-wise LightGBM against traditional baselines, restores minority sensitivity via training-isolated SMOTE, resolves regulatory opacity via local SHAP attributions, and audits economic disparate impact under SR 11-7 standards.

3. Competitor Analysis

In the financial technology and credit risk engineering landscape, "competitors" encompass both commercial enterprise scoring solutions and methodological approaches deployed across the banking industry. This analysis evaluates three primary competitor archetypes operating in this market space:

1. **Traditional Parametric Scorecards:** Generalized Linear Models / Logistic Regression (e.g., standard FICO scorecards, SAS Credit Scoring).
2. **Proprietary Commercial AutoML / Credit AI Platforms:** Enterprise black-box risk suites (e.g., Zest AI, DataRobot, Underwrite.ai).
3. **Standard Academic / Open-Source ML Benchmarks:** Naive ensemble implementations (e.g., baseline scikit-learn / XGBoost pipelines without banking compliance constraints).

3.1. Business Case and Industry Context

Commercial lending institutions operate under stringent capital requirements: every basis point of default misclassification translates into either direct charge-off losses or excessive regulatory capital provisioning under Basel III. Conversely, improperly rejecting creditworthy applicants incurs severe opportunity costs in lost net interest margin and customer lifetime value.

While enterprise AI software promises substantial reductions in non-performing loans, commercial banks are slow to adopt third-party black-box tools due to vendor lock-in, multi-million dollar licensing fees, and immense difficulty defending proprietary algorithms during regulatory exams (SR 11-7 audits). There is an immediate market demand for transparent, reproducible, in-house machine learning pipelines that can bridge high predictive discrimination with verifiable regulatory compliance.

3.2. Comparative SWOT Analysis

Competitor / Framework	Strengths (S)	Weaknesses (W)	Opportunities (O)	Threats (T)
Archetype 1: Traditional Parametric Scorecards (Logistic Regression, FICO Baselines)	• Universally accepted by global regulators (Basel III, SR 11-7). • Monotonic score behavior guaranteed. • Native parameter interpretability. • Seamless, low-cost core banking integration.	• Incapable of capturing non-linear feature interactions. • Highly sensitive to extreme financial outliers (e.g., DTI spikes). • Lower discrimination (ROC-AUC 0.575). • Severe underfitting on tail-risk defaults.	• Hybridization with machine learning feature selection. • Integration of non-traditional data via Generalized Additive Models (GAMs).	• Significant market share loss to agile FinTech lenders deploying ML. • Adverse selection: prime borrowers poached by competitors offering better terms.
Archetype 2: Enterprise Commercial AI Suites (Zest AI, DataRobot, Underwrite.ai)	• Highly polished graphical interfaces. • Automated hyperparameter tuning and model governance dashboards. • Proprietary explainability toolkits marketed directly for compliance. • Enterprise client support.	• Extremely expensive recurring annual licensing costs. • Proprietary "black-box" wrappers that limit internal quantitative auditability. • Risk of undetected target leakage in automated feature engineering. • Rigid cloud vendor lock-in.	• Expansion into tier-2 regional banks and credit unions lacking internal quantitative teams. • Partnership with core banking providers.	• Regulatory pushback against automated, non-reproducible feature pipelines. • Emergence of high-capability open-source XAI libraries such as SHAP, reducing commercial moat.
Archetype 3: Standard Academic / Kaggle ML Benchmarks (Naive GBDT, XGBoost Baselines)	• High global statistical ranking metrics (ROC-AUC > 0.80). • Open-source, flexible, and zero licensing cost. • Fast algorithmic execution using modern GBDT frameworks such as LightGBM.	• Pervasive target leakage from post-origination behavioral artifacts. • Naive 0.50 threshold collapse (0% recall on rare defaults). • Lack of regulatory governance, XAI integration, and adverse action reporting. • Resampling leakage (SMOTE applied prior to train-test splits).	• Serves as the technical foundation for custom enterprise engineering. • Rapid adoption across academic financial engineering curricula.	• Disqualification in real-world banking audits due to SR 11-7 non-compliance. • High risk of algorithmic bias and disparate impact lawsuits under ECOA.
Proposed Project Framework: Leakage-Free LightGBM + SMOTE + SHAP + Fairness Audit	• Mathematically honest: strict temporal boundary eliminating all post-origination leakage. • Regulatory aligned: granular adverse-action generation via TreeSHAP force plots. • Operationally calibrated: threshold-tuned for minority detection under 1.8% class imbalance. • Ethically audited: evaluates FPR parity across economic income proxies. • Zero software licensing cost: built entirely on open-source Python tools.	• Dataset scale constrained (10,000 observations in current iteration). • Computational latency of SHAP calculations in ultra-high-frequency streaming environments. • Modest precision (7.1%) due to extreme underlying class skew.	• Direct integration into two-stage Expected Loss (PD × LGD × EAD) portfolio frameworks. • Deployment in mid-market retail lending seeking bank-owned IP without vendor fees. • Scaling to multi-million-loan public bureau datasets such as LendingClub and Fannie Mae.	• Shifting macroeconomic regimes causing credit distribution drift and requiring retraining pipelines. • Emergence of newer deep tabular models (e.g., TabNet), though lacking equal tree explainability.

3.3. Market Positioning and Unique Strategic Contribution

This project occupies a distinct position at the intersection of quantitative rigor, regulatory compliance, and open-source accessibility. Unlike traditional scorecards (Archetype 1), this model captures non-linear interactions, achieving superior minority default detection. Unlike commercial suites (Archetype 2), this framework provides complete transparency, full algorithmic ownership, and reproducible auditability at zero software licensing cost. Unlike naive academic benchmarks (Archetype 3), this framework enforces strict leakage prevention, addresses class imbalance without evaluation corruption, provides localized adverse-action explanations, and conducts formal disparate impact testing.

4. Initial Findings, Code Highlights, and Repository Access

4.1. Online Code Repository Access

The complete codebase, exploratory notebooks, and modular pipelines are hosted in the following repository:

https://github.com/sanaurrehmanarain/credit-risk-xai-framework

4.2. Highlights of Important Code Portions

In accordance with Module 3 guidelines, the key code segments developed during this exploratory analysis phase are highlighted below.

A. Target Definition & Data Cleaning (`notebooks/01_data_cleaning.ipynb`):

Establishing the binary default definition and imputing missing variables using domain-specific heuristics:

```
# Grouping loan status into binary target: 0 = Safe, 1 = Default/Charged Off
def create_target(status):
    if status in ['Current', 'Fully Paid']:
        return 0
    else:
        return 1

df['loan_status_binary'] = df['loan_status'].apply(create_target)

# Impute joint columns (missing indicates single applicant)
df['annual_income_joint'] = df['annual_income_joint'].fillna(0)
df['debt_to_income_joint'] = df['debt_to_income_joint'].fillna(0)

# Impute delinquency history (missing indicates no prior delinquency -> safe flag 999)
df['months_since_last_delinq'] = df['months_since_last_delinq'].fillna(999)
df['months_since_90d_late'] = df['months_since_90d_late'].fillna(999)
df['emp_length'] = df['emp_length'].fillna(0)
```

B. **Temporal Boundary & Target Leakage Prevention** (`notebooks/01_data_cleaning.ipynb / 03_feature_engineering.ipynb`):

Identifying and systematically dropping variables observable only post-origination to ensure underwriting honesty:

```
# Post-origination features that cause catastrophic target leakage:
# Including these yields an artificial AUC of 0.83; purging them restores a legitimate baseline
cols_to_drop = [
    'loan_status', 'sub_grade', 'issue_month', 'balance',
    'paid_total', 'paid_principal', 'paid_interest', 'paid_late_fees'
]
df_clean = df.drop(columns=cols_to_drop)
```

C. **Exploratory Risk Ordering Validation** (`notebooks/02_exploratory_data_analysis.ipynb`):

```
# Sanity check verifying that empirical default rates scale monotonically with loan grade
default_rates = df.groupby('grade')['loan_status_binary'].mean().reset_index()
print(default_rates)
```

4.3. Initial Findings and Temporary Results

Exploratory analysis conducted across Notebooks 01 and 02 yielded three critical empirical findings that establish the scope of subsequent model development:

1. **Extreme Empirical Class Imbalance:** The target distribution contains 9,822 non-default loans (98.2%) and only 178 default events (1.8%). This confirmed that standard probability thresholds (0.50) will cause naive classifiers to predict zero defaults, requiring training-isolated SMOTE and precision-recall threshold optimization.

2. **Monotonic Risk Ordering Sanity Check:** Empirical default rates across assigned credit grades demonstrated consistent risk-scaling: Grade A (0.7%), Grade B (1.1%), Grade C (1.9%), Grade D (3.4%), Grade E (5.1%), and Grade F (10.5%). This validates that the engineered target variable captures actual credit risk patterns rather than random noise.

3. **Distributional Skew and Extreme Financial Outliers:** Key financial ratios exhibited severe positive skewness; for example, the Debt-to-Income (DTI) ratio spans from 0.0 to a maximum of 469.09 (with a median of 17.57), while Annual Income reaches $2,300,000. This empirical distribution explains why traditional linear solvers (such as Logistic Regression) struggle with convergence, providing quantitative justification for transitioning to histogram-based, leaf-wise decision trees (LightGBM) that split on feature quantiles and remain robust to extreme values.

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